

CHAPTER TWO – LITERATURE REVIEW

2.1 Introduction

Effective communication is essential in educational institutions. Students, lecturers, and academic administrators rely on timely and accurate information exchange to facilitate learning, coordinate academic activities, and ensure successful completion of academic tasks. The rapid growth of mobile technology and internet accessibility has led to the widespread adoption of digital communication platforms within universities.

This chapter reviews existing communication technologies used in academic environments, examines their strengths and limitations, and identifies the research gap that motivates the development of Campuz.

2.2 Academic Communication Systems

Academic communication systems are digital platforms that facilitate the exchange of educational information among students, lecturers, and academic administrators. These systems support the sharing of announcements, assignments, schedules, learning materials, examination information, and collaborative discussions.

An effective academic communication platform should provide reliable information delivery, easy access to academic resources, structured organization of content, and mechanisms for ensuring that important information reaches intended recipients.

With the increasing adoption of mobile devices among students, mobile-based communication systems have become a preferred medium for academic interactions due to their accessibility and convenience.

2.3 WhatsApp in Academic Communication

WhatsApp is one of the most widely used messaging applications in educational environments. Many course representatives and lecturers use WhatsApp groups to distribute announcements, share lecture materials, and communicate with students.

The popularity of WhatsApp can be attributed to its ease of use, instant messaging capabilities, multimedia support, and widespread adoption among students.

However, WhatsApp was designed primarily as a social messaging platform rather than an academic communication system. Consequently, academic information often becomes mixed with unrelated conversations, making it difficult to retrieve important announcements and resources. Students may also miss important messages due to high message volumes within groups.

Advantages of WhatsApp

- Easy to use and widely adopted.

- Supports file sharing and multimedia communication.
- Enables instant communication among users.
- Provides group communication features.

Limitations of WhatsApp

- Lack of structured academic organization.
- Important information can be buried among conversations.
- Limited search and categorization capabilities.
- No intelligent academic reminder system.
- No dedicated academic resource management.

2.4 Telegram

Telegram is another messaging platform commonly used for educational communication. It offers larger group capacities, cloud-based storage, and support for channels.

Telegram allows administrators to broadcast information to large audiences and supports file sharing without strict size limitations.

Despite these advantages, Telegram remains a general-purpose communication platform and does not provide dedicated features for managing academic schedules, assignments, deadlines, or educational resources.

Advantages of Telegram

- Supports large groups and channels.
- Cloud-based file storage.
- Flexible administrative controls.
- Supports large file uploads.

Limitations of Telegram

- Not specifically designed for academic communication.
- Limited academic workflow management.
- Lack of intelligent academic support features.

2.5 Google Classroom

Google Classroom is a learning management system designed to support educational activities. It enables lecturers to distribute assignments, manage coursework, and communicate with students.

The platform provides better organization of academic content compared to traditional messaging applications.

However, Google Classroom is primarily designed for formal classroom management and often requires institutional adoption and administrative setup. It may not provide the informal and rapid communication experience that students are accustomed to through messaging applications.

Advantages of Google Classroom

- Structured assignment management.
- Organized learning materials.
- Integration with Google services.
- Academic-focused environment.

Limitations of Google Classroom

- Requires institutional setup.
- Less suitable for informal communication.
- Limited real-time communication features.

2.6 KNUST V-Class System

KNUST V-Class is the official Learning Management System (LMS) used by Kwame Nkrumah University of Science and Technology (KNUST) for the delivery of academic content and management of course-related activities. The platform provides lecturers and students with a centralized environment for accessing learning materials, course announcements, assessments, and other academic resources.

V-Class plays an important role in supporting teaching and learning within the university by providing a structured course-based environment for academic content delivery.

Features

- Course-based organization of academic content.
- Uploading and distribution of learning materials.
- Announcements section for course updates.
- Calendar functionality for scheduled academic activities.
- Dashboard for accessing enrolled courses.

Strengths

- Official institution-controlled platform.
- Organized and structured course content delivery.
- Centralized access to academic materials.
- Integrated calendar functionality for scheduling academic activities.
- Supports academic resource management within a single environment.

Limitations

- Not designed for real-time communication.
- Limited interaction between students and lecturers.
- No chat-based communication system between students and course representatives.
- No smart deadline extraction or automated reminder functionality.
- Requires users to manually navigate through courses to access information.
- Lacks real-time push-based notification delivery; users must actively check for updates.
- Lower engagement levels compared to widely used messaging applications.

Although V-Class provides effective course management and content delivery, it does not fully address the need for fast, interactive, and real-time academic communication among students, lecturers, and course representatives. This creates an opportunity for a complementary platform that combines structured academic organization with modern communication features.

2.7 Microsoft Teams

Microsoft Teams combines communication, collaboration, and file management functionalities within a single platform. Educational institutions use Teams for online classes, meetings, announcements, and collaboration.

While Microsoft Teams provides extensive features, its complexity may present challenges for students seeking a lightweight communication solution.

Advantages of Microsoft Teams

- Comprehensive collaboration features.
- Supports meetings and video conferencing.
- Integration with Microsoft Office tools.
- Organized communication channels.

Limitations of Microsoft Teams

- Higher complexity.
- Steeper learning curve.
- Resource-intensive on some devices.

2.8 Research Gap

The review of existing systems reveals that while communication and learning management platforms provide valuable features, they do not fully address the specific communication challenges faced by university students.

General messaging applications such as WhatsApp and Telegram provide convenience but lack structured academic organization. Learning management systems such as Google Classroom and Microsoft Teams provide organization but may be complex, institution-dependent, or less suitable for rapid student communication.

While KNUST V-Class provides structured academic content delivery, it lacks real-time communication, intelligent reminders, and interactive engagement mechanisms commonly required by students for day-to-day academic coordination.

There is therefore a need for a platform that combines the simplicity and familiarity of messaging applications with structured academic communication features tailored specifically for university environments.

2.9 Proposed Solution

To address the identified challenges, this project proposes Campuz, a mobile-based academic communication platform designed specifically for students, course representatives, and lecturers.

Campuz introduces Academic Communication Hubs where academic information can be organized into categories such as announcements, assignments, resources, schedules, and discussions. The platform also incorporates intelligent features including deadline reminders, calendar synchronization, AI-assisted information retrieval, and optional SMS redundancy for urgent announcements.

By combining the familiarity of modern messaging applications with structured academic workflows, Campuz aims to improve communication efficiency, accessibility, and information management within universities.

2.9.0 Summary

This chapter reviewed existing communication platforms and learning management systems commonly used in academic environments. The review identified several limitations, including poor information organization, lack of intelligent academic support features, and inadequate mechanisms for ensuring reliable delivery of important announcements.

The identified research gap justifies the development of Campuz as a specialized academic communication platform that addresses these limitations while maintaining a familiar and accessible user experience.